

KALKULUS SECURYTAS

How Data Storage & Serverless Execution Works on Vercel

1. Executive Summary & Answer Key

"Kalkulus Securytas uses Vercel Serverless Functions with a temporary ephemeral /tmp file system for 100% temporary, self-deleting data transfers between devices."
If anyone asks you how Kalkulus Securytas stores data on Vercel, here is the exact 1-sentence answer to give them.

2. Technical Storage Architecture on Vercel

- **Serverless Micro-Containers:** Vercel executes Node.js Express routes as serverless functions. Each function runs inside an isolated micro-container that spins up instantly on demand.
- **The Ephemeral /tmp File System:** On Vercel, standard file paths are read-only. However, Vercel provides a temporary writable directory located at `/tmp`` (up to 512MB per instance).
- **File Payload Storage:** When a user drops a PowerPoint, PDF, Video, Audio, or Document file, Multer writes the uploaded file to `/tmp/uploads/``.
- **PIN & Clip Metadata Storage:** When a text clip, code snippet, or file PIN is generated, the server saves the clip object and PIN code (e.g. `4829``) into `/tmp/data_store.json``.

3. Complete Data Lifecycle (How Data Travels)

- 1. Upload & PIN Generation:** User uploads a 24MB file or code snippet from phone. Vercel serverless function receives the payload, writes the file to `/tmp/uploads/``, assigns a unique 4-digit PIN (`4829``), and saves metadata to `/tmp/data_store.json``.
- 2. Instant Cross-Device Retrieval:** Receiver opens the app on another device/browser and enters `4829``. Vercel serverless function reads `/tmp/data_store.json``, locates PIN `4829``, streams the file from `/tmp/uploads/``, and delivers the payload instantly.
- 3. Automatic Expiration & Disk Purge:** A 60-second background cleanup routine continuously checks expiration timestamps. Once the clip expires (e.g. 1 hour or 24 hours), `fs.unlinkSync`` permanently unlinks the file from disk, ensuring ZERO permanent data retention.

4. Quick Q&A Reference Sheet (How to Answer Questions)

Q: Where are files stored on Vercel?

A: In Vercel's ephemeral `/tmp/uploads`` directory attached to the serverless function environment.

Q: Is database setup (PostgreSQL / MongoDB) required?

A: No! Because Kalkulus Securytas is designed for 100% temporary transfers, using `/tmp`` JSON file persistence eliminates external database costs and latency.

Q: What happens when files expire?

A: The server automatically calls `fs.unlinkSync()`` to permanently purge the file from storage.

Q: What is Vercel's request payload limit?

A: Vercel Free (Hobby) tier allows up to 4.5MB per serverless request. For larger files (up to 200MB), Render persistent hosting can be used.